Vector Differentiation: Gradient of r^n

1. Gradient of a Power of Radius

Name of Concept

Gradient of a Radial Power Function (r^n).

Core Mathematical Formula

$$\nabla r^n = nr^{n-2}\bar{r}$$

Beginner-Friendly Explanation of Operations

1. **Leveraging a Known Identity:** Instead of calculating the partial derivatives with respect to x, y, and z from scratch every single time, we can use a previously proven shortcut: the general formula for the gradient of any radial function, $\nabla f(r) = f'(r)\frac{\bar{r}}{r}$.
2. **Applying the Power Rule:** We treat r^n like a standard single-variable calculus term. To find its derivative with respect to r, we pull the power n to the front and decrease the exponent by 1, giving us nr^{n-1}.
3. **Combining Exponents (Laws of Indices):** When we combine our derivative with the baseline formula, we end up dividing a base of r by another base of r. According to exponent rules:

$$\frac{r^{n-1}}{r^1} = r^{(n-1)-1} = r^{n-2}$$

This simple subtraction of exponents explains why the final formula features r^{n-2} instead of r^{n-1}.

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2. Step-by-Step Proof

Question

Prove that $\nabla r^n = nr^{n-2}\bar{r}$.

Step 1: Recall the standard radial gradient identity

We begin with the established identity for the gradient of any scalar function depending solely on distance r:

$$\nabla f(r) = f'(r)\frac{\bar{r}}{r}$$

Step 2: Define the specific function and find its derivative

Assign our specific function to $f(r)$:

$$f(r) = r^n$$

Differentiate $f(r)$ with respect to r using the standard calculus power rule:

$$f'(r) = nr^{n-1}$$

Step 3: Substitute the derivative back into the identity

Replace $f'(r)$ in our primary equation with the calculated power rule expression:

$$\nabla r^n = \left(nr^{n-1}\right)\frac{\bar{r}}{r}$$

Step 4: Group the scalar components together

Rearrange the terms to group all scalar operations involving r away from the position vector $\bar{r}$:

$$\nabla r^n = \left(n \cdot \frac{r^{n-1}}{r} \right) \bar{r}$$

Step 5: Simplify the exponents to reach the final form

Apply the laws of indices ($\frac{x^a}{x^b} = x^{a-b}$) to simplify the fraction:

$$\nabla r^n = \left(n \cdot r^{(n-1)-1} \right) \bar{r}$$

$$\nabla r^n = nr^{n-2}\bar{r}$$